

$$\{\text{Only 1- and 2- loops} = \begin{pmatrix} 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\omega^3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\omega^3 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\omega^3 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\omega^3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\omega^3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\omega^3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 \end{pmatrix},$$

$$\text{1- and 2- cycles along with 1- and 2- loops} = \begin{pmatrix} 2 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & -\omega^3 & 0 & 0 & 0 & \omega^2 & \omega & 0 & 0 \\ 0 & 0 & -\omega^3 & \omega & 0 & 0 & 0 & \omega^2 & 0 \\ 0 & 0 & \omega^2 & -\omega^3 & 0 & 0 & 0 & \omega & 0 \\ 1 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 \\ 0 & \omega & 0 & 0 & 0 & -\omega^3 & \omega^2 & 0 & 0 \\ 0 & \omega^2 & 0 & 0 & 0 & \omega & -\omega^3 & 0 & 0 \\ 0 & 0 & \omega & \omega^2 & 0 & 0 & 0 & -\omega^3 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 2 \end{pmatrix},$$

$$\text{All 1's and all 2's} = \begin{pmatrix} 2 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 2 \\ 0 & -\omega^3 & 0 & 0 & 0 & -\omega^3 & -\omega^3 & 0 & 0 \\ 0 & 0 & -\omega^3 & -\omega^3 & 0 & 0 & 0 & -\omega^3 & 0 \\ 0 & 0 & -\omega^3 & -\omega^3 & 0 & 0 & 0 & -\omega^3 & 0 \\ 2 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 2 \\ 0 & -\omega^3 & 0 & 0 & 0 & -\omega^3 & -\omega^3 & 0 & 0 \\ 0 & -\omega^3 & 0 & 0 & 0 & -\omega^3 & -\omega^3 & 0 & 0 \\ 0 & 0 & -\omega^3 & -\omega^3 & 0 & 0 & 0 & -\omega^3 & 0 \\ 2 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 2 \end{pmatrix},$$

$$\text{Everything except 0-loops} = \left. \begin{pmatrix} 2 & 0 & 0 & 0 & 3 & 0 & 0 & 0 & 3 \\ 0 & -\omega^3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\omega^3 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\omega^3 & 0 & 0 & 0 & 0 & 0 \\ 3 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 & 0 & -\omega^3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\omega^3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\omega^3 & 0 \\ 3 & 0 & 0 & 0 & 3 & 0 & 0 & 0 & 2 \end{pmatrix} \right\}$$